Day 092 | Unit 07 | Teacher plan

Understand and Use Percent

Relate Fractions, Decimals, and Percents

Standards: 6.RP.A.3c

Learning target: I can connect a percent with a fraction out of 100 and a decimal.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100.

Worked example 1

Write 50% as a simplified fraction and decimal.

1. Percent means per 100.

2. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal.

3. 50% = 50/100 = 1/2

4. divide by 100 for 0.5.

5. State the result in the form requested: 1/2; 0.5.

Answer: 1/2; 0.5

Worked example 2

Select all representations of 60%.
A. 60
B. 0.6
C. 60/10
D. 60/100

1. Percent means per 100.

2. Percent means per 100.

3. Divide 60 by 100 to get 0.6

4. simplify 60/100 to 3/5.

Answer: B, D - 0.6; 60/100

Day 092 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write 25% as a simplified fraction and decimal.

02 / Constructed response

Write 20% as a simplified fraction and decimal.

03 / Constructed response

Write 10% as a simplified fraction and decimal.

04 / Constructed response

Write 5% as a simplified fraction and decimal.

05 / Constructed response

Write 60% as a simplified fraction and decimal.

06 / Constructed response

Write 75% as a simplified fraction and decimal.

07 / Constructed response

Write 30% as a simplified fraction and decimal.

08 / Multiple select

Select all representations of 40%.
A. 0.4
B. 40/100
C. 40
D. 40/10

09 / Multiple select

Select all representations of 350%.
A. 350/10
B. 350
C. 3.5
D. 350/100

10 / Constructed response

Write 15% as a simplified fraction and decimal.

Day 092 | Unit 07 | Answer key

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1. 1/4; 0.25

25% = 25/100 = 1/4; divide by 100 for 0.25.

6.RP.A.3c | Constructed response

2. 1/5; 0.2

20% = 20/100 = 1/5; divide by 100 for 0.2.

6.RP.A.3c | Constructed response

3. 1/10; 0.1

10% = 10/100 = 1/10; divide by 100 for 0.1.

6.RP.A.3c | Constructed response

4. 1/20; 0.05

5% = 5/100 = 1/20; divide by 100 for 0.05.

6.RP.A.3c | Constructed response

5. 3/5; 0.6

60% = 60/100 = 3/5; divide by 100 for 0.6.

6.RP.A.3c | Constructed response

6. 3/4; 0.75

75% = 75/100 = 3/4; divide by 100 for 0.75.

6.RP.A.3c | Constructed response

7. 3/10; 0.3

30% = 30/100 = 3/10; divide by 100 for 0.3.

6.RP.A.3c | Constructed response

8. A, B - 0.4; 40/100

Percent means per 100. Divide 40 by 100 to get 0.4; simplify 40/100 to 2/5.

6.RP.A.3c | Multiple select

9. C, D - 3.5; 350/100

Percent means per 100. Divide 350 by 100 to get 3.5; simplify 350/100 to 7/2.

6.RP.A.3c | Multiple select

10. 3/20; 0.15

15% = 15/100 = 3/20; divide by 100 for 0.15.

6.RP.A.3c | Constructed response